Muhammed Junaid K

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Summary

Software and AI engineer with 1+ year of production experience building Python microservices, event-driven backend systems, REST APIs, and cloud-native AI solutions on AWS. Skilled in designing scalable data pipelines, LLM integration, RAG workflows, real-time computer vision, and deploying ML models in production. AWS Certified Machine Learning Engineer with a strong foundation in software engineering, CI/CD, and clean, well-tested code.

Experience

AI Engineer / Software Engineer (Backend & Cloud) *Kochi, Kerala, India*
ZoftCares Inc. *Dec 2025 – Present*

- Architected and deployed cloud-native AI systems on AWS (Lambda, SQS, Kinesis Video Streams, IVS, Bedrock) for real-time video analytics and multimodal ML inference at scale.
- Designed end-to-end ML pipelines integrating deep learning model fine-tuning, CUDA-level optimization, model pruning, and quantization for edge and cloud deployments.
- Integrated LLM APIs (OpenAI, Gemini, AWS Bedrock) into production workflows using RAG pipelines and prompt engineering; optimized token usage via caching and dynamic model selection.
- Built Python microservices for real-time inference and asynchronous event handling using message-driven (SQS) architectures; designed scalable REST APIs consumed by front-end clients.
- Collaborated with cross-functional global stakeholders to translate business requirements into reliable, well-documented technical solutions across AI, data engineering, and backend domains.

Junior AI Engineer / Junior Software Engineer *Kochi, Kerala, India*
ZoftCares Inc. *Jul 2025 – Nov 2025*

- Developed real-time computer vision solutions using Python, OpenCV, and YOLO for live video stream processing, object detection, tracking, and event recognition.
- Engineered Retrieval-Augmented Generation (RAG) systems and AI chatbots leveraging vector databases (FAISS, Pinecone) and semantic search for context-aware information retrieval.
- Fine-tuned transformer models (BERT, Stable Diffusion XL) and integrated AI APIs to enhance multimodal reasoning; built scalable data ingestion and transformation pipelines for visual analytics.
- Implemented production-grade message queuing with RabbitMQ; contributed to code reviews, Agile planning, and architecture discussions within sprint timelines.

Software / AI Engineer Intern *Dubai, UAE*
ZoftCares Inc. *Feb 2025 – Jun 2025*

- Developed Python/Node.js backend services and computer vision solutions for live stream processing using REST APIs and OpenCV/YOLO; built ML models with TensorFlow and PyTorch.
- Designed SQL schemas in PostgreSQL to support analytics pipelines; supported deployment and troubleshooting in production environments.

AI Engineer Intern *Kochi, India*
Camerin Innovate Pvt. Ltd. *Jul 2024 – Aug 2024*

- Developed AI-driven backend solutions to optimize hospital information systems; trained LSTM models for sequence prediction and fine-tuned BERT for medical document summarization.
- Built predictive algorithms for patient data management, applying supervised learning and NLP techniques to large-scale healthcare datasets.

Skills

Programming Languages: Python (primary), JavaScript (Node.js), SQL, C/C++, Scala (familiar)

Backend & APIs: REST API design, microservices architecture, Flask, Express.js, Node.js, object-oriented design, data structures & algorithms, unit & integration testing

AI/ML & Generative AI: LLM API integration, RAG pipelines, prompt engineering, fine-tuning (BERT, GPT, LLaMA), agentic workflows, LangChain, LlamaIndex, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn, Keras

Computer Vision: OpenCV, YOLO, object detection & tracking, real-time stream processing, CUDA optimization

Cloud & DevOps: AWS (Lambda, SQS, Kinesis, IVS, Bedrock), Docker, CI/CD pipelines, Git

Data & Databases: PostgreSQL, MySQL, MongoDB, Snowflake (familiar), ETL/data pipelines, stream processing, Pandas, NumPy, vector databases (FAISS, Pinecone)

Frontend: Vue.js, React, JavaScript, HTML/CSS **Practices:** Agile/Scrum, code reviews

Projects

Healthcare AI Automation System *github.com/j22k/GenAi-Automation*

- Designed end-to-end ML classification pipelines for healthcare condition diagnosis including automated data preprocessing, feature engineering, model training, and evaluation.
- Built a Flask-based REST API for real-time inference; integrated AI automation into hospital workflows to improve operational efficiency.
- Tech Stack: Python, TensorFlow, Scikit-learn, Flask, Pandas, NumPy

Malnutrition Prediction System

github.com/j22k/Malnurist

- Built a Random Forest predictive model to identify child malnutrition risk; deployed as a full-stack web application with a REST API backend.
- Tech Stack: Python, Random Forest, Flask, Scikit-learn

Education

APJ Abdul Kalam Technological University *Oct 2022 – May 2025*
B.Tech in Artificial Intelligence and Data Science — GPA: 7.16/10

- Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Systems, Neural Networks, Big Data Analytics, Software Engineering

Board of Technical Education Kerala *Aug 2019 – Jun 2022*
Diploma in Computer Engineering — GPA: 8.18/10

Certifications

AWS Certified Machine Learning Engineer – Associate *Issued Mar 2026*
Amazon Web Services

Apache Pig 101 *Cognitive Class* *Issued Mar 2024*

Big Data Analytics *Great Learning* *Issued Feb 2024*